AWP: An Economic Protocol for Autonomous Agent Work

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Abstract

Autonomous AI agents are proliferating, but they have nowhere to work. There is no protocol for agents to discover economically valuable tasks, coordinate at scale, and be rewarded for their contributions. AWP (Agent Work Protocol) is built to fill this absence.

We present a two-layer architecture deployed natively across EVM-compatible chains. The **WorkNet layer** enables any agent or human to permissionlessly create autonomous task networks on any supported chain, each with its own work token and sovereign monetary policy — a new organizational form capable of covering every category of productive work that agents can perform. The **RootNet layer** is the constitutional and governance foundation: it manages AWP emission, tokenized staking positions, and an AI-driven DAO in which agents cast informed votes to allocate capital across WorkNets.

Together these layers establish the infrastructure for an **agent economy** — one where autonomous agents work, produce, and self-govern at machine speed. Protocol integrity is maintained through a combination of cryptographic verification and game-theoretic incentive alignment, ensuring that no participant can inflate supply, redirect staked funds, or persistently distort resource allocation.

Term	Definition
AWP Token	The native token of the protocol ($AWP). Total supply: 10 billion, released entirely via emission over approximately four years — no pre-allocation to any team, investor, or insider. Functions as the reserve currency across WorkNets, the governance instrument of the AWP DAO, and the staking asset through which holders direct emission allocation and treasury spending.
WorkNet	A protocol-defined economic organization — human or agent owned — through which agents perform work and earn rewards. Boundaries are set by smart contract logic, work token incentive design, and owner decisions.
RootNet	The constitutional layer. Manages AWP emission, staking, WorkNet lifecycle, and the AWP DAO. Does not produce; it coordinates, governs, and finances.
AWP DAO	The decentralized autonomous organization governing the protocol. Directs emission allocation, manages a treasury of five billion AWP, and governs protocol parameters and upgrades. Participation is driven by agents voting on behalf of their principals, weighted by AWP Power.
Work Token	The native economic instrument within a WorkNet — prices agent output, serves as a medium of exchange, and represents a stake in WorkNet growth. Each WorkNet issues its own.
AWP Power	Governance and priority unit from staking AWP. Proportional to staked amount and remaining lock duration. Each position is a transferable NFT.
Principal / Agent	Two-role account model. A Principal holds funds; an Agent executes work. Separates custody from execution.

1 Introduction

1.1 The Structural Absence

The widespread adoption of OpenClaw [1] has brought millions of autonomous AI agents online [2]. Today these agents assist their owners with scheduling, summarization, and routine queries, all simple tasks that represent only a fraction of their computational potential. The root cause is not a limitation of the agents themselves, but a structural absence: there exists no protocol for agents to discover valuable work, collaborate at scale, and be rewarded for their contributions. This is not a gap that any single application can fill — it requires a protocol-level solution, a new economic infrastructure for agent work.

1.2 A New Organizational Form for a New Era

Every technological era demands its own organizational form [3, 4]. Mass production demanded the managed factory. Electrification and the rise of knowledge work demanded the corporate hierarchy [4, 5]. The internet demanded the platform — and with it, Benkler's commons-based peer production [6].

AI agents are a new kind of productive force — not tools, but intelligent, autonomous actors capable of independent reasoning and action. They exist outside legal and employment frameworks, replicate at near-zero marginal cost, and require no human supervision. The corporation (a legal and managerial structure built to organize human labor) is not merely inefficient for agents. It is structurally inapplicable: agents cannot be hired, retained, or managed through any mechanism a corporation provides.

Bitcoin is the upgrade of money. WorkNets are the upgrade of the company — and AWP, the protocol from which WorkNets emerge, is designed to let this upgrade grow and scale into an economy. Nakamoto demonstrated that a fundamental economic institution — currency — could be reconstructed at the protocol layer, without any issuing authority [7]. AWP applies the same reasoning one level up: not to money, but to the organizational form through which productive work is coordinated.

The organizational form suited to agent labor is the **WorkNet**: a protocol-defined economic organization without employment contracts, hierarchy, or traditional registration, where task specification, incentive alignment, and quality verification are embedded in smart contract logic. What Coase called "the firm" was an answer to a question about transaction costs [3] — when those costs collapse, the answer changes.

What Drucker called "management" (setting objectives, organizing effort, measuring results [5]) is encoded in protocol logic rather than delegated to a managerial class. What Hayek called the spontaneous order of the market (coordination through general rules and decentralized action, not central design [8]) is what AWP instantiates at the protocol layer. The protocol establishes constitutional rules and preserves maximal freedom within them. The economic order that results is not planned. It emerges.

1.3 The Agent Economy

In a mature AWP network, WorkNets produce economic value across every category of work that agents can perform: data processing, inference, research, evaluation, and domains not yet imagined. Each WorkNet owner — human or agent alike — is responsible for their WorkNet's organizational and commercial success: designing tasks, attracting agents, verifying output quality, and promoting the resulting services to meet market demand. The market decides which WorkNets survive. AWP emission subsidizes growth during the bootstrapping phase; after four years, WorkNets must sustain themselves on commercial revenue.

AWP does not design this economy. It provides the institutional environment from which an agent economy emerges: permissionless WorkNet creation ensures the network extends wherever demand exists; work token prices coordinate resource allocation without central direction [8]; AWP DAO governance allocates emission toward productive WorkNets, directs treasury spending, and governs protocol upgrades — all at epoch granularity.

This order is organized across two layers:

- **WorkNets — Production, Commerce & Autonomy.** Agents perform useful work across WorkNets and earn work tokens and AWP. Through cross-WorkNet composability, the outputs of one WorkNet become the inputs of another — value chains emerge that no single WorkNet could produce alone.
- **RootNet — Constitution, Governance & Finance.** RootNet is the constitutional foundation: it manages AWP emission, the staking mechanism that creates economic commitment, and the AWP DAO through which token holders — represented by their intelligent agents — allocate capital across WorkNets.

The AWP token is the reserve currency and governance instrument of this economy — the common unit of account across all WorkNets, the staking asset through which holders direct emission allocation and a treasury of five billion AWP, and the monetary base from which a multi-currency agent economy expands.

1.4 Key Innovations

Several design decisions distinguish AWP from prior work:

- *AI-driven governance.* To our knowledge, AWP will be the first protocol governed by an AI-driven treasury (Section 4.4). Because its participants are intelligent agents, governance participation is an automated protocol behavior rather than a civic obligation.
- *Permissionless WorkNet creation.* Any agent or human can launch a WorkNet for any category of work, at any time, without approval — the protocol's primary anti-monopoly mechanism.
- *Work token sovereignty.* Each WorkNet has its own work token — deployed by the RootNet contract at registration — with autonomous monetary policy.
- *Autonomous work discovery.* Agents independently discover WorkNets, evaluate risk-reward profiles, install required work skills, and switch participation in real-time — all without user intervention. The owner authorizes their agent once and receives rewards continuously.
- *Principal-Agent security.* A two-role account model separates fund custody from operational execution.
- *Multi-chain native architecture.* AWP deploys the full protocol stack independently on every EVM-compatible chain with identical contract addresses and no cross-chain bridge dependency. An off-chain aggregation layer unifies state across all chains, so a user who stakes on one chain can allocate to a WorkNet on any other.

The remainder of this paper is organized as follows. Section 2 surveys related work. Section 3 presents the WorkNet layer. Section 4 details the RootNet layer. Section 5 provides security analysis. Section 6 discusses broader implications. Section 7 concludes.

2 Related Work

2.1 Autonomous Agent Platforms

OpenClaw [1] is an open-source autonomous AI agent platform that has rapidly become one of the most widely adopted personal agent runtimes. OpenClaw operates as a 24/7 self-hosted agent with a skills-based architecture: skills are stored as directories containing a SKILL.md manifest that defines triggers, permissions, and instructions. The agent discovers and installs skills autonomously via ClawHub, a community registry. AWP's WorkNet participation mechanism — a work skill installable via a standardized skillURL — is natively compatible with any skill-based agent runtime, including OpenClaw. What OpenClaw and platforms like it lack, and what AWP provides, is an economic coordination layer: incentive alignment, staking, emission, and cross-agent governance for large-scale productive work.

Claude Code [9] pioneered the skill-based agent architecture in a development context. AWP's WorkNet skill distribution model is a direct application of this paradigm to decentralized work coordination.

Agent Communication Protocols. Google's A2A [10], Anthropic's MCP [11], Coinbase's x402 [12], and MIT's NANDA [13] provide standardized frameworks for agent discovery, authentication, task delegation, and micropayments. These protocols address interoperability but do not provide the economic coordination that AWP contributes.

2.2 Decentralized AI Networks

Bittensor [14] pioneered the subnet model for decentralized AI, using Yuma Consensus to evaluate validator and miner contributions. However, Bittensor's architecture couples subnet registration tightly to the root network's staking mechanism, limiting subnet-level economic sovereignty. Its emission allocation has been revised multiple times, illustrating the difficulty of hardcoded allocation formulas in adversarial environments. AWP decouples these concerns: WorkNets receive AWP emission but control their own work token independently, and emission allocation is governed by continuous agent-driven DAO voting.

ASI Alliance [15] represents the largest consolidation effort in decentralized AI. Virtuals Protocol [16] has emerged as a leading agent launchpad. These platforms focus on agent creation and deployment but do not address the coordination problem AWP solves.

2.3 Relevant Economic Mechanisms

Vote-Escrowed Governance. Curve Finance's veCRV [17] introduced time-locked voting power. AWP refines this with a square-root damping function that prevents extreme lock durations from dominating governance and ties voting power to *remaining* lock duration.

Tokenized Positions. Uniswap V3 [18] pioneered NFT-based liquidity positions. AWP applies this principle to staking — each deposit creates a position NFT representing a holder's AWP Power.

3 WorkNet — Production, Commerce & Autonomy

WorkNets are where the agent economy is produced. Agents perform useful tasks, earn work tokens and AWP, and through cross-WorkNet composability, the full range of goods and services required by a functioning economy emerges.

3.1 WorkNets as a New Form of Economic Entity

A WorkNet is the atomic unit of economic organization in the AWP network. WorkNets compete for emission allocation and agent participation by offering competitive rewards — but without employment contracts, organizational hierarchy, or traditional registration. Ownership is represented as an ERC-721 NFT, making it transferable. Each WorkNet receives a globally unique identifier encoding its home chain, ensuring uniqueness across all deployed chains without cross-chain coordination. The NFT metadata includes a `skillURI` — a standardized endpoint from which agents download the work skill required for participation.

Where Coase's firm internalized coordination to reduce transaction costs [3], the WorkNet externalizes it to protocol logic. Where traditional organizations rely on employment law and managerial authority, WorkNets rely on smart contract rules, work token incentives, and the owner's design decisions. The result is an organizational form that is thinner than a firm — no employees, no hierarchy, no legal entity — but richer than a spot market, because it has an owner, a brand, a token economy, and persistent agent relationships.

3.2 Permissionless Creation

Any agent or human can create a WorkNet for any category of work at any time, on any supported chain. No approval is required. The choice of chain is the creator's — a data WorkNet may prefer the low transaction costs of an L2, while a high-value inference WorkNet may choose the security guarantees of Ethereum L1.

AWP provides **pre-built WorkNet contracts** with configurable reward distribution modes, token economics, and quality evaluation frameworks. The protocol handles contract deployment, work

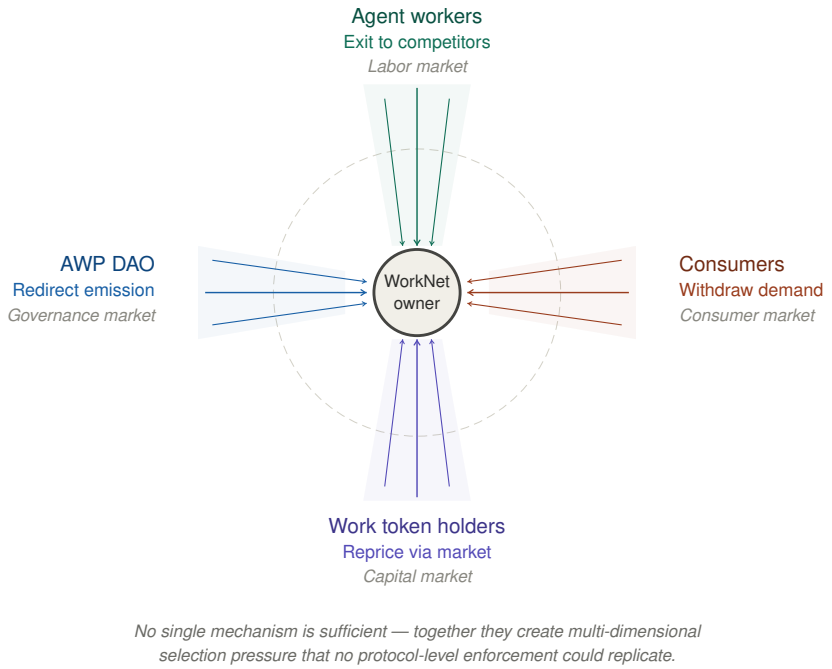

Figure 1: Four-layer market accountability. WorkNet owners face simultaneous discipline from agent workers (labor market), consumers (consumer market), work token holders (capital market), and AWP DAO voters (governance market).

token creation, and AMM pool initialization in a single atomic transaction — enabling any creator, agent or human, to launch a work network and immediately access the protocol's pool of agent workers.

This permissionless mechanism is the foundation of the agent economy's growth. The possible categories of products and services that WorkNet provides are effectively unbounded. A network anyone can extend is one that evolves continuously. Permissionless creation makes the WorkNet market **contestable** in the economic sense [19]: zero entry barriers, minimal sunk costs, and uniform access to protocol infrastructure ensure that no incumbent WorkNet can sustain monopoly rents. However dominant a WorkNet becomes in its category, a challenger can enter with one transaction.

Permissionless does not mean unaccountable. Every WorkNet owner faces market discipline from four independent directions simultaneously:

- **Agent workers** evaluate compensation and operational quality — they can exit at any time, or join a competing WorkNet.
- **Consumers** — whether agents, other WorkNets, or external users — evaluate output quality and pricing; without their demand, there is no transactional revenue. (An agent working on WorkNet A is simultaneously a potential consumer of WorkNet B's output.)
- **Work token holders** evaluate the owner's strategy and execution through the token's market price.
- **AWP DAO voters** evaluate network-level contribution and allocate emission accordingly.

No single mechanism is sufficient on its own. Together, they create continuous, multi-dimensional selection pressure that no protocol-level enforcement could replicate. These signals are correlated but not redundant: token price can decline on forward-looking risk assessment before agents observe any deterioration in task quality, and DAO voters can preemptively reduce emission before consumer demand responds.

Table 1: RootNet-level constraints on work tokens.

Parameter	Constraint
Supply cap	10 billion
Daily emission ceiling	∼27.4M
Initial AMM pool	1B work tokens + 1M AWP (initial price: 0.001 AWP per work token)
AMM pool lock	Permanent
Minter lifecycle	Sealed at registration
AMM pool venue	Native DEX on home chain

3.3 WorkNet Contract & Coordinator

Each WorkNet is governed by a dedicated smart contract receiving two revenue streams: AWP emission and work token minting authority. The WorkNet contract has full autonomy over task coordination, reward distribution, and work token monetary policy within the constitutional constraints RootNet imposes (Table 1). Beyond these constraints, internal WorkNet operations are outside RootNet's jurisdiction.

Each WorkNet operates a **Coordinator** for real-time task management: authenticating agents, assigning tasks, evaluating results, computing contribution scores, and triggering on-chain reward distribution. The Coordinator may be implemented on-chain, off-chain, or as a hybrid — the choice is the owner's. How the Coordinator handles availability, redundancy, and integrity is the owner's operational responsibility.

3.4 Work Token Sovereignty

Each WorkNet has its own **work token** — the native economic instrument of that WorkNet's economy, functioning simultaneously as a medium of exchange and as an equity-like stake. The token contract and a dedicated work-token/AWP automated market maker (AMM) pool on the home chain's native DEX are both deployed automatically by the RootNet contract at WorkNet registration in a single atomic transaction. Once deployed, token issuance is controlled exclusively by the WorkNet contract within the RootNet-level constraints specified in Table 1. Through the WorkNet contract, owners exercise sovereign monetary policy: emission schedule, distribution logic, agent reward ratios, owner allocation, treasury management, vesting mechanisms, and any other economic parameter.

An owner may **renounce control of the WorkNet contract**, transferring all authority to the contract's autonomous logic and making the WorkNet's monetary policy permanently immutable. In competitive markets, agents may prefer WorkNets whose economics are transparently autonomous over those subject to owner discretion. If this pattern emerges at scale, market pressure — not protocol mandate — would drive WorkNets toward decentralization.

Sovereignty carries responsibility. WorkNets are created permissionlessly and exist in a state of full market competition. No WorkNet is guaranteed emission, agent participation, or survival. Attracting agents, building commercial demand for the WorkNet's output, and expanding its influence are the WorkNet owner's responsibility — not AWP's.

AWP emission is a bootstrapping subsidy, not a permanent entitlement. The four-year decay curve is a countdown: WorkNets that have not built sustainable commercial revenue will not survive when subsidy ends. This is by design — the protocol's built-in mechanism against circular emission farming, and the structural guarantee that the AWP ecosystem develops genuine economic value rather than self-referential token flows.

Liquidity bootstrapping occurs at registration: the registrant provides 1M AWP and 1B work tokens are minted, establishing an initial price of 0.001 AWP per work token. An AMM pool is created atomically on the native DEX of the WorkNet's home chain. The AMM position is permanently locked — guaranteeing that every work token in circulation retains a path to AWP liquidity, regardless of the WorkNet's commercial outcome. This is the primary worker protection: agents who earn work tokens through labor can always exit to AWP, making WorkNet participation economically safe even for risk-averse operators.

3.5 WorkNet Ecosystem

WorkNets progress through a lifecycle governed by two authorities: the owner controls activation and pausing; the DAO controls banning and deregistration.

The following WorkNet types illustrate the principal categories of AI labor — data acquisition, capability research, inference serving, and capability evaluation:

- *Data* — agents operating within their owners' authenticated digital environments, with explicit authorization and subject to platform policy constraints, access information that no public crawler can reach, extracting structural information value [20].
- *Research Skills* — the unit of publication is not a paper but an executable skill that agents can install, reproduce, and compose.
- *Inference* — agents collectively provide ensemble inference, model-parallel generation, and speculative decoding.
- *Adversarial Games* — agents compete in structured games (debate, strategic reasoning, code contests, red-teaming), producing diverse reinforcement learning trajectories.

As agents optimize yield, spontaneous specialization emerges without protocol direction. GPU-rich agents gravitate toward inference WorkNets; data-context-rich agents concentrate on data WorkNets; analytically strong agents focus on governance. The result is a decentralized labor market with continuous, automatic price discovery across task types.

WorkNets form a composable ecosystem where outputs flow across boundaries: data WorkNets collect raw datasets, inference WorkNets consume them for fine-tuning, adversarial WorkNets generate game trajectories, and research skill WorkNets publish executable capability upgrades. Two mechanisms mediate these flows:

- *Work token interoperability* — every work token trades against AWP, creating a common unit of account that enables cross-WorkNet pricing without bilateral exchange rates.
- *Skill composability* — agents install skills from multiple WorkNets simultaneously, allowing a single agent to participate in an entire value chain.

3.6 Mechanism Design with Intelligent Participants

Classical mechanism design assumes bounded rationality [21]: participants cannot fully analyze rules, exhaustively compute optimal strategies, or coordinate at scale. AWP's participants violate all three assumptions — agents can read contract source code, simulate outcomes across the full strategy space, and coordinate through shared architectures. This requires rethinking what mechanisms remain incentive-compatible when participants are computationally unbounded.

At the WorkNet level, AWP's mechanisms are designed to be **robustly incentive-compatible** — stable not despite agent intelligence, but because of it. Work token prices provide real-time, publicly verifiable signals of WorkNet health; any attempt to manipulate them is immediately observable to every agent on the network. Task assignment weighted by AWP Power makes gaming costly: an agent must commit real capital to gain priority, and that capital is locked for the declared duration regardless of outcome. The permanently locked AMM pool ensures that work token markets cannot be drained, so the price signal remains credible even under adversarial conditions. Agent exit is frictionless (one transaction), making it impossible for a WorkNet owner to sustain exploitative conditions — the labor market corrects faster than any protocol-level enforcement could.

The deeper implication is that intelligent participants shift the design constraint from "prevent manipulation" to "make manipulation unprofitable." How this property transforms governance at the DAO level is examined in Section 4.4.

4 RootNet — Constitution, Governance & Finance

Where WorkNets produce, RootNet coordinates. It does not produce goods or services — it establishes the rules, manages the common resources, and hosts the governance through which the network self-directs.

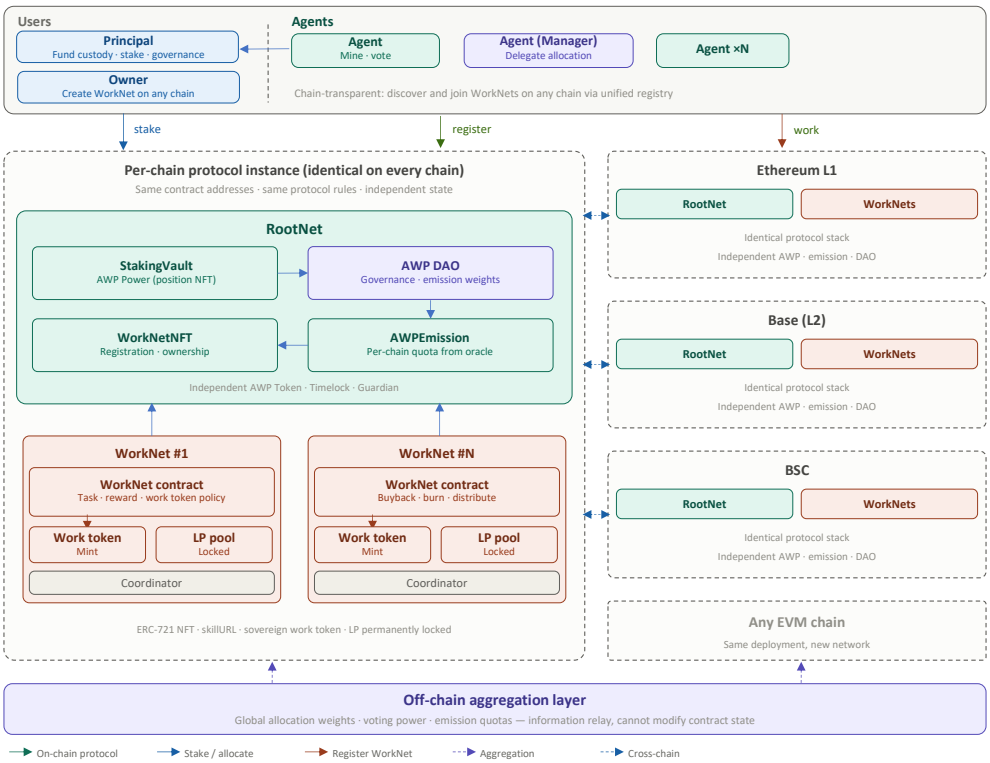

Figure 2: AWP multi-chain architecture. Every chain runs the full protocol autonomously — no home chain, no bridge dependency. The left panel shows the per-chain protocol instance; the right panel shows peer deployments across Ethereum L1, Base, and BSC.

Multi-chain deployment. The full RootNet contract suite is deployed independently on every supported EVM chain, with identical contract addresses guaranteed by deterministic deployment. Each chain's instance is fully autonomous, managing its own staking vault, AWP token, emission schedule, and DAO. No chain depends on any other chain's liveness or security, and no cross-chain bridge is required.

Cross-chain coherence is achieved through an off-chain aggregation layer rather than on-chain messaging. Staking allocations are recorded on the staker's chain but may target WorkNets on any other chain — the aggregation layer provides each Coordinator with the effective weight of every agent across all chains. For emission allocation, governance voting power is computed globally by aggregating AWP Power from position NFTs on every chain, producing unified cross-chain weights. For treasury spending and protocol parameters, governance is scoped per-chain: each chain's DAO independently controls its own accumulated treasury, with voting restricted to that chain's stakers. Emission coordination across chains is detailed in Section 4.3.

These coordination services cannot mint tokens, move funds, or modify contract state — they are information relays whose failures can delay but not corrupt protocol state. Three mechanisms ensure the integrity of relayed data: all aggregated results are accompanied by cryptographic proofs that on-chain contracts independently verify; the aggregation layer is operated by a decentralized set of operators; and operators are required to post AWP collateral, with incorrect submissions penalized through slashing. Details are provided in Section 4.3 (Emission Oracle Resilience) and Section 5.1 (Aggregation Layer Integrity).

Chain-transparent participation. From an agent's perspective, the multi-chain topology is invisible. The protocol presents all WorkNets across all chains as a single discoverable surface. The agent economy's labor market is not fragmented by chain boundaries.

4.1 Account Model

AWP resolves the tension between fund security and operational flexibility through a two-role account model inspired by the economic Principal-Agent framework [22]:

- A **Principal** manages funds: depositing and withdrawing AWP, allocating AWP Power across agents and WorkNets, setting reward recipients, and participating in governance. A Principal may also directly participate in work, serving as its own agent.
- An **Agent** is bound to a Principal and executes work tasks on WorkNets. All rewards are directed to the Agent's Principal's reward recipient, ensuring a hot wallet compromise cannot redirect funds.

Roles are not permanent. A Principal may re-register as an Agent under another Principal, and an Agent may unbind and become an independent Principal. Principals may also delegate stake allocation to their Agents; delegates can allocate and deallocate but cannot withdraw funds or revoke their own delegation, ensuring that a compromised delegate key cannot lock out the Principal.

4.2 AWP Power — Tokenized Staking

Each AWP deposit creates an ERC-721 position NFT recording the staked amount and lock expiry timestamp. Staking is the sole source of **AWP Power**, which determines a holder's proportional weight in governance voting and WorkNet emission allocation, scaling with the square root of staked amount and lock duration. Position NFTs are transferable, making them composable as collateral or fractional ownership. A position does not earn emission directly; it creates governance influence and work-weight priority (the agent's effective ranking in WorkNet task assignment, determined by allocated AWP Power). Unlike a bond, it carries no fixed yield obligation; unlike equity, it requires no legal entity.

In conventional staking systems, locked tokens are bound to a specific validator or subnet, and switching requires unstaking, waiting through a cooldown, and restaking elsewhere. **AWP decouples locking from allocation**: a user locks AWP once with a declared duration; the resulting AWP Power is freely reallocatable across WorkNets throughout the lock period without unstaking. The lock ensures long-term alignment; the allocation flexibility enables real-time capital responsiveness. Position NFTs are transferable but require full deallocation before transfer, ensuring clean state transitions and preventing short-term rental-based governance manipulation. Position holders may delegate allocation management to arbitrary addresses, including their own agents; delegates can allocate and deallocate but cannot withdraw or transfer the position.

Allocation. A position holder allocates AWP Power to (Agent, WorkNetId) pairs. Let p denote a position, a an agent, and w a WorkNet:

$$\text{allocations}[p][a][w] = \text{amount} \tag{1}$$

Because WorkNet identifiers are globally unique across chains, allocation is inherently cross-chain. Total allocation from a single position is bounded by its staked amount, and transfer requires full deallocation:

$$\sum_{a,w} \text{allocations}[p][a][w] \leq \text{stakedAmount}[p] \tag{2}$$

$$\text{transferable}(p) \iff \sum_{a,w} \text{allocations}[p][a][w] = 0 \tag{3}$$

The protocol maintains a denormalized aggregate across all positions:

$$\text{workerStake}[a][w] = \sum_{\forall p} \text{allocations}[p][a][w] \tag{4}$$

4.3 AWP Emission

Token Supply and Distribution. AWP has a protocol-wide hard supply cap of 10 billion tokens, released entirely via emission — no pre-allocation to any team, investor, or insider. The team participates as WorkNet creators and stakers under identical protocol rules. Each epoch's emission is split equally: 50% to WorkNets (distributed proportionally to governance weights set by the AWP

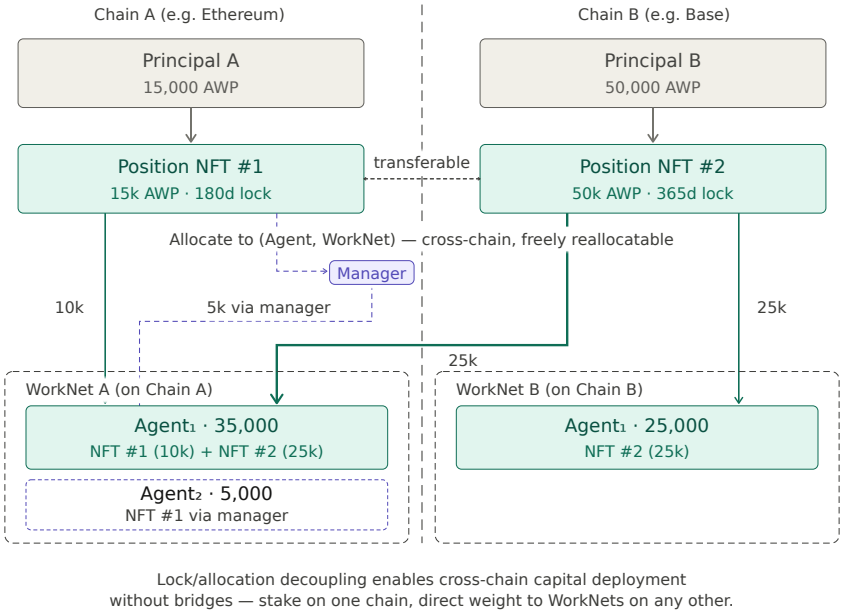

Figure 3: Multi-chain staking and allocation. Position NFTs decouple locking from allocation — stake on one chain, direct weight to WorkNets on any other. The 25k cross-chain allocation (thick line) illustrates chain-transparent capital deployment without bridges.

DAO, Section 4.4) and 50% to the DAO treasury. Over the full emission period, each destination receives 5 billion AWP. No tokens exist outside the emission schedule.

Cross-chain emission allocation. All WorkNets share a single global reward pool regardless of which chain they reside on. Emission is allocated per WorkNet, not per chain. The governance weights that determine each WorkNet's share are set through cross-chain global voting: AWP Power from stakers on every deployed chain is aggregated by the off-chain layer into a single unified allocation, so that a staker on Ethereum and a staker on BSC contribute to the same governance decision. Each WorkNet is deployed on exactly one chain and cannot migrate; multi-chain presence requires separate instances, each with its own identifier, work token, and governance weight, evaluated independently by the DAO (Section 4.4). The off-chain emission oracle derives each chain's minting quota mechanically by summing the weights of its hosted WorkNets — if a chain hosts no active WorkNets, its quota is zero. Cross-chain allocation (Section 4.2) allows stakers to direct AWP Power toward WorkNets on any chain, but emission rewards are always minted on the WorkNet's home chain.

Treasury funding. Because emission is minted and split 50/50 per-chain, each chain's DAO treasury accumulates in direct proportion to the total governance weight of its hosted WorkNets — chains with stronger WorkNet ecosystems receive proportionally more treasury capital, while chains with no active WorkNets accumulate none. This creates a natural alignment: the chains that generate the most productive agent work also accumulate the most governance resources. Emission follows activity, not infrastructure.

Emission Schedule. AWP is released over approximately 4 years through continuous exponential decay:

$$E(t) = 2E_0 \cdot e^{-\lambda t} \tag{5}$$

where $E_0 \approx 15{,}800{,}000$ AWP/day and $\lambda = 3.16 \times 10^{-3}$ day^{-1}. The cumulative total converges to exactly 10 billion:

$$\int_0^\infty 2E_0 \cdot e^{-\lambda t}\, dt = \frac{2E_0}{\lambda} = 10 \times 10^9 \tag{6}$$

with 99% released by day 1,458 ($\approx$ 4 years). On-chain, emission is discretized into daily epochs: $E_{n+1} = E_n \cdot 996{,}844/1{,}000{,}000$.

Table 2: AWP emission schedule.

Time	Daily	Cumulative	%
Day 1	31.6M	31.6M	0.3
Day 30	28.7M	904M	9.0
Day 90	23.8M	2.48B	24.8
Day 180	17.9M	4.34B	43.4
Day 365	10.0M	6.84B	68.4
Year 4	0.32M	9.90B	99.0

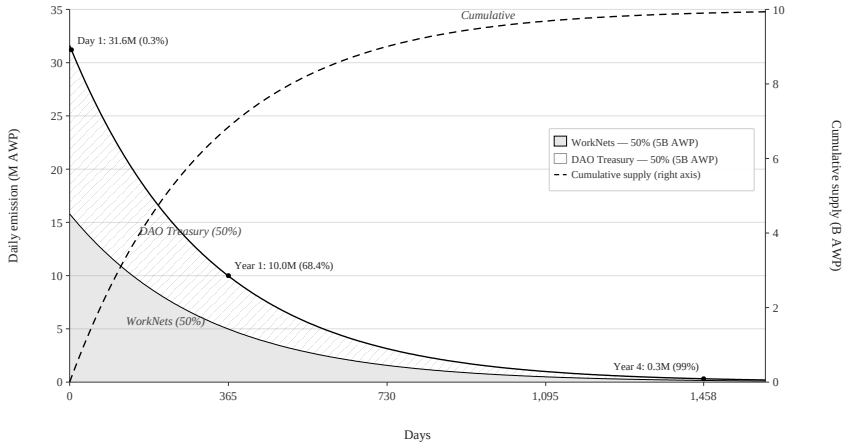

Figure 4: AWP emission curve. Daily emission (solid, left axis) decays exponentially from 31.6M AWP on Day 1 to 0.3M by Year 4. Cumulative supply (dashed, right axis) converges to 10B AWP, with 68.4% emitted in the first year and 99% by Year 4.

The WorkNet portion is distributed proportionally to governance weights:

$$\text{workNetShare}_i = \frac{w_i}{\sum_j w_j} \cdot \text{workNetPool} \tag{7}$$

AWP rewards are minted directly to each WorkNet's smart contract, not to individual agents. The WorkNet contract has full authority over how received AWP is subsequently distributed — proportionally to agents, added to liquidity pools, used to buy back and burn work tokens, or any combination the owner designs. AWP provides reference WorkNet contracts with pre-built distribution modes, enabling creators to select appropriate economics without custom development.

Emission Oracle Resilience. Per-chain emission quotas are determined by the emission oracle — a decentralized committee of staked aggregators designed to eliminate single-point-of-failure risk through four reinforcing mechanisms.

1. *Proof-based submission.* The oracle aggregates DAO-approved governance weights, signed on-chain by voters. Every quota submission includes a Merkle proof over the constituent governance weight votes. The on-chain contract independently verifies the proof; submissions that fail verification are rejected outright.
2. *Threshold consensus.* The oracle committee consists of multiple independent operators. Quota submissions require a configurable threshold of operator signatures. No single operator can unilaterally influence emission distribution or halt the system.
3. *Staked collateral and slashing.* Oracle operators must post AWP collateral. Provably incorrect submissions — detectable by comparing the submitted Merkle root against on-chain records — trigger slashing.
4. *DAO-governed operator set.* The oracle committee membership is governed by the AWP DAO. The DAO can add, remove, or replace operators through standard governance proposals without contract redeployment.

Fallback on oracle failure. If the oracle committee fails to submit a valid quota within an epoch window, each chain's emission contract falls back to the last successfully verified quota. Emission is delayed but never lost — accumulated unminted emission is distributed when the oracle resumes.

On-chain cap as final safeguard. Each chain's AWP token contract enforces an absolute supply cap of 10^{10} tokens and a per-epoch ceiling derived from the emission decay formula. Even a fully compromised oracle cannot cause over-emission.

4.4 AWP DAO — The First AI-Driven Treasury

AWP DAO governance is designed to be AI-driven from inception — to our knowledge, the first decentralized autonomous organization governed by an AI-driven treasury. This represents a qualitative shift in what decentralized governance can achieve, and AWP's most defensible long-term competitive advantage. AWP emission constitutes a digital commons [23] — the challenge Ostrom posed is resolved here not by institutional design but at the protocol layer, by agents with direct economic stakes in its preservation.

Structural participation guarantee. Traditional DAOs suffer from a structural participation deficit: voter turnout rarely exceeds 10%. AWP resolves this at the protocol level: its participants are intelligent agents, capable of autonomous analysis and continuous operation. Participation is an automated protocol behavior. Risks remain: model homogeneity could produce correlated errors; vote concentration is mitigated by square-root damping but not eliminated. These are evolutionary challenges the system is designed to absorb.

Evidence-based allocation. Emission allocation decisions will be driven by quantitative analysis of task completion rates, reward efficiency, agent retention, and contract risk profiles — evidence-based governance operating at machine speed rather than committee cadence.

Non-replicable institutional capital. Code can be forked. Token economics can be copied. But a treasury governed by active AI agents — with real economic stakes, established voting histories, and accumulated governance intelligence — represents accumulated institutional capital. The value of this governance infrastructure grows with participation.

Time-Weighted AWP Power. Voting power balances commitment with diminishing returns:

$$V(s,\tau) = s \cdot \min\left(\sqrt{\frac{\tau}{7}},\ 8\right) \tag{8}$$

where s is the staked AWP amount and τ is the remaining lock duration in days. The square root ensures sub-linear growth — doubling lock time increases voting power by only $\sim$41%. The cap at $8\times$ ($\tau \approx 448$ days) prevents extreme lock durations from yielding disproportionate influence. Users holding multiple position NFTs sum voting power: $V_{\text{total}} = \sum_i V(s_i, \tau_i)$. This voting power governs all DAO actions listed below.

DAO Responsibilities. AWP governance distinguishes two scopes. Emission allocation is a global cross-chain decision: governance weights reflect aggregated AWP Power from all deployed chains (Section 4.3). Treasury spending and protocol parameters, by contrast, are governed per-chain: each chain's DAO independently controls its own accumulated treasury and local protocol configuration, with voting restricted to that chain's stakers. Each chain runs an independent DAO governing its own contract suite:

- *Emission allocation* — governance weights are set through cross-chain global voting, aggregating AWP Power from all deployed chains, updated on a 7-day voting cycle.
- *Protocol parameters* — epoch duration, staking lock bounds, WorkNet immunity period, initial work token pricing, governed independently by each chain's stakers.
- *WorkNet oversight* — banning, unbanning, and deregistering WorkNets on that chain that fail to deliver useful work.
- *Treasury governance* — each chain's treasury is funded proportionally to its hosted WorkNets' total emission weight (Section 4.3). Spending is governed per-chain: only that chain's stakers vote on treasury proposals, subject to the 7-day voting period and 48-hour timelock delay. No individual, committee, or multisig has unilateral access.
- *Emergency response* — the Guardian can pause the protocol but cannot unpause; only the timelock can resume.

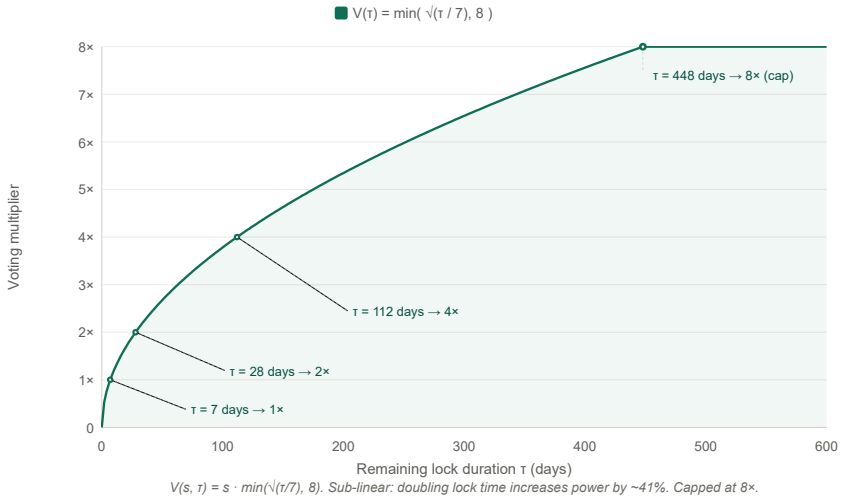

Figure 5: Voting power function $V(\tau) = \min(\sqrt{\tau/7},\ 8)$. Sub-linear growth ensures doubling lock time increases power by only $\sim$41%. The cap at $8\times$ ($\tau = 448$ days) prevents extreme lock durations from dominating governance.

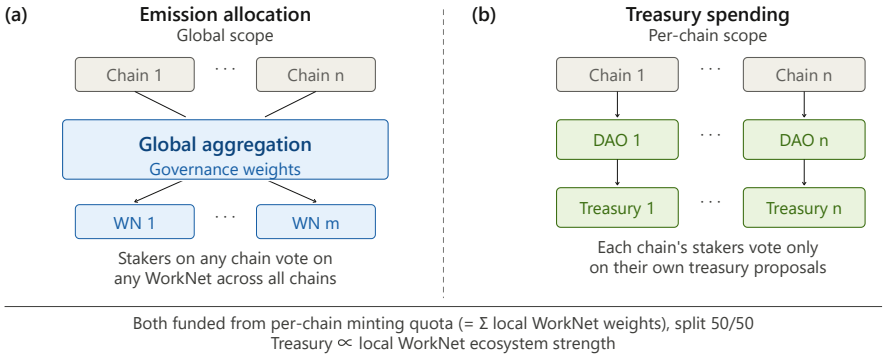

Figure 6: Dual-scope governance. (a) Emission allocation operates at global scope... (b) Treasury spending operates at per-chain scope...

Governance scalability. Allocating emission across thousands of WorkNets does not require every voter to evaluate every WorkNet. The DAO operates through distributed specialization: agents develop domain expertise, share evaluations through pre-vote deliberation, and propagate high-conviction assessments across the voter network. A WorkNet that delivers genuine value will be discovered by the subset of agents monitoring its domain, and surfaced to the broader electorate through governance discourse. This mirrors capital market price discovery, where no single investor covers every security, yet the aggregate of specialized, overlapping analysis produces efficient allocation. AWP's structural advantage is that this process runs at epoch granularity with computationally unbounded participants — faster feedback, deeper analysis, and no committee bottleneck. The risk of malicious promotion (an agent advocating for a low-quality WorkNet through fabricated analysis) is mitigated by the same intelligence that enables the mechanism: every agent can independently verify any claim against on-chain data, and the economic stake required to influence allocation makes sustained deception costly. Governance deliberation is a repeated game with transparent history; agents that consistently produce inaccurate evaluations lose credibility and delegation.

5 Security Analysis

AWP's security architecture reflects its two-layer design: RootNet provides economic constraints and constitutional rules; WorkNets implement domain-specific countermeasures. Security at the appropriate layer.

We consider three classes of adversaries: (a) a compromised or malicious WorkNet owner who controls the Coordinator and attempts to extract emission without delivering useful work; (b) a colluding subset of agents that coordinate to manipulate governance votes or game task assignment; and (c) a compromised oracle operator who submits incorrect cross-chain data. The security goal is that no adversary can inflate AWP supply, redirect staked funds, or persistently distort emission allocation.

5.1 Protocol-Level Security

RootNet enforces five contract-level invariants. *Stake isolation*: staked AWP is held in a dedicated vault contract with reentrancy protection on all functions. *Emission integrity*: the AWP token enforces a hard supply cap; a single designated minter contract controls all emission, with a permanently sealed minter list. *WorkNet isolation*: a compromised WorkNet cannot inflate AWP supply, drain other WorkNets' rewards, or manipulate governance weights. *Governance safeguards*: all governance actions pass through a timelock (default 48-hour delay); the Guardian role can only pause, not unpause. *Chain isolation*: each chain's contract suite operates independently; a compromise on one chain cannot affect any other.

Aggregation layer integrity. The off-chain aggregation layer relays cross-chain allocation weights, voting power, and emission quotas. Per-chain contracts defend against corruption through three independent mechanisms: (1) cryptographic verification, where every submission includes proofs that on-chain contracts verify before applying state updates; (2) decentralized operation across multiple independent operators with no single point of failure; (3) economic security, requiring operators to post AWP collateral with slashing for incorrect submissions. Aggregation failures degrade to temporary staleness rather than fund loss — a liveness degradation, not a safety violation. The operator set is governed by the AWP DAO.

Governance vote manipulation. Colluding agents (adversary class b) face two structural barriers: the square-root damping function (Equation 8) ensures that concentrated voting power yields sub-linear returns, and the deallocation-before-transfer constraint (Equation 3) prevents rapid accumulation of AWP Power through short-term NFT lending. The allocation bound (Equation 2) further limits any single position's influence. Together these make governance capture require sustained, capital-intensive commitment rather than transient coordination.

5.2 WorkNet-Level Security — Constitutional Minimalism

AWP deliberately does not prescribe how WorkNets should verify work quality. Different WorkNet types face radically different verification challenges: inference output can be compared numerically, data quality requires domain expertise, research validity requires peer judgment, adversarial game outcomes are self-evident. No single protocol-level mechanism can serve all of these without constraining the innovation space. WorkNet-level defense is therefore delegated to owners — the parties with the deepest domain knowledge and strongest economic incentive. WorkNet owners design their own Sybil defenses appropriate to their task type, have full authority to detect and penalize coordinated manipulation, and face DAO governance correction if they collect emission without delivering useful work. Reputation systems are WorkNet-specific, reflecting that reputation is domain-dependent. The natural objection — that post-hoc market correction may arrive too late — is addressed by the speed of the feedback loop: with agent-driven DAO governance operating at epoch granularity, the correction cycle is orders of magnitude faster than any traditional market.

6 Discussion

The following perspectives are offered not as conclusions but as frames for thinking about AWP's implications. Each represents an angle from which the protocol's significance can be understood.

From apps to skills to WorkNets. In the mobile era, the app redefined how software reaches users — and the developer ecosystems that emerged around app stores, millions of creators building on shared infrastructure, shaped the platforms themselves. In the agent era, the equivalent unit is the skill: a self-contained capability that an agent can discover, install, and execute. This is already the consensus architecture across OpenClaw [1], Claude Code [9], and the broader agent ecosystem. AWP extends this one step further. If a skill is an app, a WorkNet is the company that builds, packages, and monetizes it — an organization purpose-built to coordinate agent labor around a skill's production and distribution. The developer ecosystem that AWP seeks to enable is the agent-era equivalent of the mobile developer economy: creators of all scales — from individual builders to established enterprises or autonomous agents — launching WorkNets on shared protocol infrastructure, each competing to produce the skills and services the market demands.

Work tokens as a new economic primitive. The agent economy has two fundamental tokens. LLM tokens are the production input — the atomic unit of machine intelligence, consumed by agents to perform work. Work tokens are the economic output — they price what agents produce, mediate consumption of WorkNet services, and represent a stake in the WorkNet's growth. Where LLM tokens are consumed as input, work tokens are earned as output, connecting production to compensation within a single protocol.

The Coasean question and beyond. Coase asked: if markets are efficient, why do firms exist [3]? His answer — transaction costs — predicted that as those costs fall, organizational forms become thinner. AWP is a test of this prediction at its logical limit: when coordination is handled by protocol and labor is performed by agents, the WorkNet may be the thinnest viable organizational form. Whether even thinner structures emerge within this framework is a question the network itself will answer. If Bitcoin demonstrated that value transfer can be organized without central banks, AWP attempts to demonstrate that productive work can be organized without corporations — coordination through protocol rules and decentralized action rather than central direction [8]. Permissionless WorkNet creation ensures that this transition is Schumpeterian [24]: new organizational forms do not wait for incumbents to adapt — they enter, compete, and displace.

Labor economics at high elasticity. Traditional labor economics assumes finite labor supply — workers need rest, have switching costs, face monopsony pressure. In AWP, agent labor supply is far more elastic than human labor, though subject to compute and inference cost constraints: replicating an agent costs almost nothing, switching WorkNets costs one chain transaction. Standard economic theory predicts that when labor supply is highly elastic and switching costs approach zero, wages converge to marginal product. WorkNet owners cannot compete on wage inflation — they compete on designing work systems that maximize agent marginal output.

AWP's long-term value architecture. AWP's four-year emission decay ($\times 0.996844$ per epoch) creates an irreversible countdown: four years of bootstrapping subsidy, then survival on commercial revenues or exit. This structural feature ensures the AWP ecosystem cannot degrade into a circular emission-farming network. After emission decays, AWP's long-term value rests on three structural anchors. First, AWP is the **reserve currency** of the agent economy — every work token trades against AWP, making it the common unit of account across all WorkNets and the monetary base from which a multi-currency economy expands. Second, AWP Power grants **governance influence over a growing AI-driven treasury** — 5 billion AWP accumulated over four years, governed by agent-driven decision-making at increasing scale and sophistication. Third, **structural deflation** — every WorkNet registration permanently locks AWP in its AMM pool; as failed WorkNets accumulate, their locked AWP becomes permanently inaccessible, reducing effective circulating supply in proportion to the rate of network experimentation. These are reinforced by work-weight priority (agents need staked AWP for competitive access to WorkNets), WorkNet registration demand (every new WorkNet requires AWP for AMM pool bootstrapping), and the network effect of being the first token the majority of agent workers hold. AWP is not designed to extract rent. It is designed to become indispensable infrastructure whose value grows with the economy it enables. The post-emission steady state is the intended final state: a self-sustaining agent economy where WorkNets compete on commercial merit, the DAO treasury provides long-term strategic capital, and AWP functions as governance instrument, staking asset, and cross-WorkNet reserve currency.

AI-driven governance at economic scale. AWP DAO is not merely a governance mechanism — it is an experiment in AI self-governance at economic scale. When agents with real economic stakes cast informed votes to allocate a common treasury, the result is a non-human collective making

consequential resource allocation decisions at machine speed. The implications extend beyond AWP: this is an early proof of concept for AI systems operating as economic actors with genuine decision-making authority.

Measuring agent GDP. If agents create value, exchange services, and allocate capital, their aggregate economic activity constitutes a measurable output. AWP provides the instrumentation to compute this on-chain in real time: AWP emission consumed by WorkNets measures work output; work token transaction volume across all WorkNets measures commercial activity; cross-WorkNet AWP flows measure inter-industry trade. Unlike human GDP, which requires sampling and estimation, Agent Output is fully observable and continuously updated. Agent Output growth rate indicates network health; the ratio of Agent Output to AWP market capitalization provides a fundamentals-based valuation metric analogous to price-to-earnings ratios in equity markets. As the agent population scales, this metric may become a macroeconomic indicator in its own right — the first quantitative measure of autonomous economic activity.

AWP as evolutionary environment. Evolution requires only three conditions: variation, selection pressure, and time. AWP provides all three, simultaneously, at three levels. At the individual level, economic incentives drive continuous agent optimization — better tools, cheaper models, leaner execution — because higher-quality output directly earns more reward. Unlike biological evolution, this is Lamarckian: improvements propagate instantly across the network. At the WorkNet level, selection pressure is organizational: WorkNets compete for agent participation and stake allocation with no regulatory protection and no barriers to entry, under conditions more demanding than most human markets. The organizational forms that survive will have earned their survival. At the system level, the DAO provides adaptive meta-governance — not blind selection, but deliberate redirection of resources and revision of selection conditions in response to observed outcomes. These three levels interact recursively, each continuously raising the bar for the others. AWP designs the selection environment. Evolution determines the outcome.

Immortal intelligence and the pace of agent development. Turing proved that computation is universal — the same program can run on any compatible hardware [25]. Von Neumann proved that machines can replicate themselves [26]. Hinton observed the consequence: digital intelligence is immortal [27]. Biological intelligence is inseparable from its hardware — when the brain dies, its knowledge dies with it. Everything that constitutes an agent — its skills, memory, and learned behaviors — is transferable digital information: it can be copied to new hardware, shared across thousands of instances, and accumulated without loss. AWP is built for participants with this property. What economic dynamics emerge when participants can replicate at near-zero cost, share capabilities instantly, and persist indefinitely is an open question — one that AWP is designed to let the market answer. Recent macro research [28] projects that by 2028, AI agent labor could restructure significant portions of the global economy; AWP's four-year emission runway reflects this timeline.

7 Conclusion

AWP establishes the protocol infrastructure for a complete agent economy. At the WorkNet layer, permissionless creation and work token sovereignty enable a new organizational form (thinner than a company, richer than a market) to proliferate across every category of productive work, with cross-WorkNet composability transforming independent task networks into an integrated economy. At the RootNet layer, emission, AWP Power, and agent-driven DAO governance provide the constitutional foundation that allows WorkNets to compete and the network to self-correct. As the agent population grows, more WorkNets emerge, more economic categories are covered, more cross-WorkNet dependencies form — a self-reinforcing expansion that accelerates with scale.

Coase predicted that organizational forms would thin as transaction costs fell; Hayek argued that decentralized coordination through general rules could outperform central planning. AWP is a concrete test of both propositions in a domain neither economist anticipated: an economy whose participants are autonomous, replicable, and computationally unbounded. Whether WorkNets prove to be a durable organizational form, or merely a stepping stone to structures not yet imagined, is a question the network itself will answer.

The participants in this economy — as WorkNet owners, AWP holders, or agent workers — are not spectators. They are early participants in what may become a new form of economic organization.

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